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APPLIED RESEARCH

Novel LSTM-Based Approaches for Enhancing Outdoor Localization Accuracy in 4G Networks

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ABSTRACT Fingerprint localization using neural networks is emerging as the state-of-the-art technique for outdoor localization using mobile network features. In this paper, we introduce two sequence-based frameworks showing major accuracy enhancements in large-scale outdoor environments compared to the state of the art in this domain. The first uses a uni-directional LSTM network called SeqOutLoc, and the second uses a bi-directional LSTM network called BiOutLoc. We also introduce the AngleNoiseSynth augmenter to expand the dataset, taking into account the angle of user movement and system noise. For SeqOutLoc, we show how adding sequence information enhances the accuracy of localization in a large-scale outdoor urban area of 45 km^2 by 25% compared to previous work, while using 35% fewer network parameters. The second model, BiOutLoc, enhances the localization accuracy with fewer network parameters by utilizing both past and future information, which is useful in near-real-time localization. To the best of our knowledge, our work is the first to use a Bi-LSTM model in outdoor fingerprint-based localization. BiOutLoc achieves a median localization accuracy of 9.4 meters, surpassing other deep learning-based localization systems by 31%, while reducing the number of parameters by 67%. Finally, we use transfer learning to fine-tune the parameters of BiOutLoc trained in a certain area using the data from a new area. This results in an 18% enhancement in accuracy and a 71% reduction in training time compared to training the model using only the data of the new area.

INDEX TERMS Data augmentation, LSTM, Bi-LSTM, outdoor localization, deep learning, transfer learning, 4G mobile communication.

I. INTRODUCTION

Accurate outdoor localization becomes essential for mobile tech-based organizations. Precise localization of mobile users helps operators direct their investments and prioritize radio optimization activities [1]. This has become even more significant in light of the expanding network connectivity driven by the proliferation of IoT [2] and widespread adoption of 4G technology [3]. According to Ericsson's mobility report published in June 2023, 4G mobile subscriptions continue to increase during Q1 2023 to 5.2 billion representing 61.2% of mobile subscriptions [4].

Even with the deployment of 5G, 4G technology will continue to play a vital role in the 5G standardization process. Early 5G networks, known as non-standalone 5G, rely on a 4G control plane to manage 5G sessions. Earlier efforts in outdoor localization problem were designed to work on 3G systems [5], [6], [7]. On the other hand, few ones are designed to work on 4G [8], [9]. In our paper, we used 4G as it represents the most used technology nowadays compared to other legacy ones or even the newer ones, 5G and 6G. Essentially, our methodology can be applied to any mobile technology, like 5G or 6G; the difference will be in the features' ranges related to each technology and any new feature specifically related to it, such as beamforming parameters in 5G and carrier phase in 6G [10], yet the system modeling will remain the same.

Outdoor localization methods in literature fall into six categories. Global Navigation Satellite Systems (GNSS) rely on satellites but face obstacles and high battery usage [11].

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Cell tower triangulation [8] determines location based on tower signals, its accuracy influenced by signal strength and tower density. Intrinsic localization in 3GPP networks faces operator reluctance due to added costs [12], [13]. Wi-Fi Positioning System relies on Wi-Fi network density [14]. Smartphone sensor-based methods [15], [16] are power-efficient but sensitive to environmental changes, limited to high-end mobiles. Fingerprint using Physical Cell Identification (PCI) in 4G networks [17]. We used this technique because it outperforms others in outdoor environments with high scattering levels and non-line of sight dominance. Additionally, it can be extended to geo-locate drive test data and geo-tag offline measurement reports saved in cloud servers at mobile operators. Typically, mobile operators use drive test data since they cannot directly evaluate the correctness of a measurement report as they do not know its ground truths, i.e., the actual mobile location in this measurement sample.

Expanding the classification perspective, outdoor localization methods are further distinguished into single-point and sequence-based techniques based on the time dimension. Single-point techniques estimate location based on a single or small number of measurements at a single point [18]. Sequence-based techniques estimate location based on a series of measurements over time [19], [20], [21], [22]. We utilized sequence-based techniques as they take into account temporal correlations between sequential positions and consider the dynamics of the device's movement and changing environment, making them more accurate than single-point techniques.

In this paper, we present two sequence-based frameworks for multiple point outdoor localization in wireless networks. The first framework, SeqOutLoc, is based on a uni-directional LSTM network, while the second, BiOutLoc, is an efficient approach that utilizes a bi-directional LSTM-based framework. Both frameworks leverage the sequence-based nature of LSTM models, and take into account temporal correlations between sequential positions. Yet extending the BiLSTM model to a localization problem is a novel approach, as usually it is utilized and proven to have good results in natural language processing (NLP) [23]. SeqOutLoc considers only past positions, while BiOutLoc utilizes both past and future information. These systems can be employed by mobile network operators and utilize various mobile network features to achieve high localization accuracy in real-time as well as near-real-time use cases where there is some flexibility in the timing of data processing and response. While the response time is still relatively fast, it may not need to be instantaneous. Examples of near-real-time use cases in telecommunications might include location-based advertising or customer localization analysis.

The main contributions of our work are summarized below:

- Our work introduces uni-directional and bi-directional LSTM deep learning models, leveraging temporal characteristics from the dataset utilized in DeepFeat [35]. These models exhibit superior accuracy, outperforming existing state-of-the-art models.
- To the best of our knowledge, our work is the first to use a Bi-LSTM model in outdoor localization. We demonstrated the advantage of bidirectional training in BiOutLoc compared to SeqOutLoc. The bi-directional LSTM model utilizes both previous and future data, providing additional information that contributes to improving localization accuracy, training time, and minimizing the number of parameters. Such a model has practical applications in near-real-time localization for mobile network operators.
- We introduce a novel data augmentation technique called AngleNoiseSynth. This technique generates synthetic samples based on system noise, user movement angles, and direction. This innovative technique showcases heightened robustness when compared to current state-of-the-art approaches.
- Our sequence-based models boast a reduced parameter count and faster convergence compared to state-of-the-art feed-forward neural networks while achieving better accuracy. This efficiency translates to lower computational demands and decreased memory allocation costs.
- Our work outperforms existing approaches in the large-scale outdoor environments. Our work achieved localization accuracy less than 10 meters in a large-scale area of 45km^2.
- We demonstrate the effectiveness of transfer learning by applying the final weights of a trained model in an urban area as the initial weights for a Bi-LSTM model in another rural area. Our approach significantly improves the overall performance of the system.

The rest of the paper is organized as follows: Section II presents previous related work. Section III presents an overview of the system. Section IV gives the details of SeqOutLoc and BiOutLoc systems. We then present the evaluation of both systems' performance in Section V. Finally, Section VI concludes the paper.

Notation: we will use the acronyms in Table1 throughout the paper.

II. RELATED WORK

This section discusses related work that addresses outdoor localization in telecom networks. Recently, significant progress has been made in the field of indoor and outdoor localization. Different models have been introduced to enhance the accuracy using different localization techniques. In the context of outdoor localization techniques, a variety of systems have been employed, each leveraging distinct mobile wireless signals, data sets, and machine learning models [5], [8], [18], [24], [25], [26]. Each system used one or more mobile features to achieve better localization accuracy. In [9], Reference Signal Received Power (RSRP) and Received Signal Strength Indicator (RSSI) measurements from drive

TABLE 1. Table of abbreviations.

Abbreviation	Explanation
IoT	Internet of Things
LSTM	Long short-term memory
GNSS	Global Navigation Satellite Systems
GPS	Global Positioning System
LoRa	Long Range
IQR	Inter-Quartile Range
MAE	Median Absolute Error
DNN	Deep Neural Network
FNN	FeedForward Neural Network
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
TA	Time Advance
RSRP	Reference Signal Received Power
RSSI	Received Signal Strength Indicator
CSI	Channel State Information
CQI	Channel Quality Indicator
PCI	Physical Cell Identifier
SINR	Signal-to-Interference-and-Noise Ratio
RSRQ	Reference Signal Received Quality
TDOA	Time Difference Of Arrival

test data have been used to geo-tag mobile measurement samples with a random Forest-based regression model in outdoor environments. On the other hand, Timing advance (TA) parameter and neighbour cell Identifiers (CELLIDs) have been utilised during handovers to detect User Equipment (UE) locations [8]. Channel state information (CSI) has been used instead of RSSI or RSRP as fingerprint data for DNN models in [27]. Aside from the diversity in the mobile features used, different machine learning models are utilised to achieve better results. PRNET+ [28] presents a multi-task learning framework integrating Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), and two attention mechanisms to effectively capture local, short-, and long-term spatial and temporal dependencies in learning measurement report features. Similarly, the DeepLoc approach [29] employs both CNN and recursive LSTM layers with measurement report data, building upon the data sets used in [28]. To address the challenges associated with noise in GPS and RSS data, data augmentation techniques [5] have been used to capture typical channel characteristics using an adequate number of samples, which in turn enhances the accuracy of Deep Neural Network (DNN) models. Beyond traditional mobile cell phone signals, Dejavu [15] utilises mobile sensors to achieve precise and energy-efficient outdoor localization. This method employs a dead-reckoning technique based on low-energy inertial sensors (e.g., accelerometer and compass) to predict user locations and address potential errors due to high noise levels in mobile sensors by integrating road landmark signal signatures to reset accumulated errors in calculations. Additionally, LoRaLoc [30] leverages Time Difference Of Arrival (TDOA) measurements generated by a LoRa network, combined with GPS location as ground truth, providing a comprehensive solution to outdoor localization challenges. In [31], A simultaneous localization and mapping (SLAM) scheme called BVLI-SLAM developed based on binocular vision, 2D lidar, and an inertial measurement unit (IMU) sensor. Most of the previous localization techniques have the following concerns regarding their practical implementation, efficient power and memory allocation, and accuracy from a mobile network operator's point of view. First, most of them use RSSI and CID only as input features, overlooking other vital telecom features such as RSRQ, TA, CQI, etc. Secondly, work that depends on base station locations need pre-identification for physical serving site coordinates, which is unavailable in different use cases. Finally, to the best of our knowledge, there is no prior work that uses a sequential model in a large-scale outdoor environment. Only [29] utilizes it in a small-scale environment (0.57 km^2) with an accuracy of 16.4m.

Table 2 provides a detailed comparison of various outdoor localization systems, highlighting aspects such as:

- The used features: Technology features as RSSI, RSRP and CID.
- Data source: The source of collected samples is either a drive test in the field or a measurement report from customer data records (CDR) from the operator's side.
- Model: The model used in the system
- Median error: Median absolute error in meters
- Area size and type: area size in km^2 and its type (urban or rural)

Comparison shows that most of the state-of-the-art systems use only two features for the localization problem. These features are the Physical Cell ID (PCI) and the RSSI.

III. SYSTEM OVERVIEW

A. LTE FEATURES

LTE is a 4G wireless standard that offers higher network capacity and user speed than 3G technology, with peak data rates of up to 100 Mbps downlink and 30 Mbps uplink speeds. It provides lower latency, scalable bandwidth capacity, and inter-technology compatibility with GSM and UMTS. Here are some of the main features and measurements of LTE [37]. First, the Physical cell ID (PCI), which is a unique identifier that is assigned to each cell in a cellular 4G network. Second one is the Reference Signal Received Power (RSRP), which measures the 4G coverage from mobile side. Typically, UE calculates RSRP for a specific cell at a given location by averaging the received power of multiple resource elements used to transfer the reference signal within the measured frequency bandwidth. RSRP provides a lot of information in one value because it is affected by several aspects as:

- Distance: RSRP indirectly reflects the distance between the device and the serving cell. As a device moves away from the cell, the received signal weakens, and the RSRP value decreases. This information can be used to estimate the device's location.

TABLE 2. Comparison between different outdoor localization systems based on the used features, model name, area size, nature, and localization accuracy. The comparison shows that most state-of-the-art systems use mainly two features (RSSI and CID) in small-scale areas.

Model	Features	Data Source	Open source dataset	Algorithm	Samples	MAE in meters	Area km^2	Type
CCR [18]	Multiple features and up to 7 neighbors RSSI	Measurement Report	No	Random Forest based regression	4749150	80	132	Urban
DeepLoc [5]	RSSI,CID	Drive test	No	DNN (4Dense)	19369	18.8	0.2	Urban
DeepLoc [5]	RSSI,CID	Drive test	No	DNN (4Dense)	44659	15.7	1.2	Rural
LUMD [9]	RSRP,RSSI	Measurement Report	No	Random Forest based regression	129000	20	4	Urban
SLN-FN [27]	CSI based fingerprint (channel state information)	Drive test	No	DNN (4Dense)	-	19.9	0.0702	Urban
PRNET [28]	RSSI,CID	Drive test	No	Hierarchical neural network	10455	13.7	0.57	Urban
TLoc [32]	RSSI,CID	Drive test	No	Random Forest based regression	7755	18.9	0.57	Urban
DeepLoc [29]	RSSI,CID	Drive test	No	CNN with Recursive LSTM	10372	16.4	0.57	Urban
RF [18], [24]	RSSI,CID, Lat/Long	Measurement Report	No	Random Forest based regression	46641	31.8	8	Urban
MLP [24]	RSSI,CID, Lat/Long	Measurement Report	No	Multilayer perceptron, simple FNN	46641	35.7	8	Urban
XGBoost [24], [33]	RSSI,CID, Lat/Long	Measurement Report	No	ML algorithms under the parallel Gradient Boosting framework	46641	41.3	8	Urban
ML [24], [34]	RSSI,CID, Lat/Long	Measurement Report	No	Maximum likelihood estimator	46641	174.6	8	Urban
Cellsense [25]	RSSI,CID	Drive test	No	probabilistic RSSI-based fingerprinting	4329	27.86	5.45	Urban
DeepFeat [35]	12 Mobile Features	Drive test	No	Maximum likelihood estimator	29375	13.7	45	Urban
LocUNet [36]	RSS, Pathloss	RadioMap	Yes	CNN	99	13.14	0.0655	Urban

- Obstacles and interference: RSRP is also impacted by obstacles such as buildings, trees, and hills, which block or weaken the signal. Interference from other signals or sources of interference (e.g., other cell towers, Wi-Fi networks) can interfere with the signal and affect the RSRP.
- Frequency band: Different frequency bands have different characteristics and impacts on RSRP. For example, higher frequency bands (e.g., 5 GHz) can have higher attenuation (weakening) due to obstacles, while lower frequency bands (e.g., 700 MHz) can have better penetration through obstacles.
- Network configuration: The design and configuration of the cellular network, including factors like cell density and cell spacing, can influence RSRP values. A network with close-spaced, dense cells will result in better RSRP.

Accordingly, RSRP for serving and neighbour cells will be one of the SeqOutLoc and BiOutLoc models inputs.

The third feature is the channel quality. There are several metrics that can be used to measure the quality of a 4G (LTE) channel:

- SINR (Signal-to-Interference-and-Noise Ratio): This is a measure of the signal-to-noise ratio (SNR) of the received signal. It considers not only the signal strength but also the interference from other signals and the background noise. A higher SINR indicates better signal quality.

$$SINR = 10 \log_{10}\left(\frac{S}{I+N}\right). \tag{1}$$

where S is the power of measured desired signals, I is the average interference power measured from other cells, and N represents the noise power.

- RSRQ (Reference Signal Received Quality): This is a measure of the quality of the downlink reference signals that are received by the device. It is calculated as the ratio of the RSRP to the total power of the received signals. A higher RSRQ indicates better signal quality.

$$RSRQ = 10 \log_{10}\left(\frac{N_B \times RSRP}{RSSI}\right). \tag{2}$$

where, RSSI represents the measured average total received power, recorded only on reference symbols, over N_B resource blocks.

- CQI (Channel Quality Indicator): This is a measure of the quality of the downlink channel that is used

TABLE 3. Table of notations.

Symbol	Description
P	Number of augmented samples in the direction of the angle towards the previous sample.
N	Number of augmented samples in the direction of the angle towards to the next sample.
T	Total number of augmented samples
D	Distance between original sample and corresponding augmented sample.
J	Number of features used as input to the deep neural network model.
Ls	Number of training samples.
F_i	Input feature to the model, where $1 \leq i \leq J$.
X	Number of grid cells in the area of interest.
GCj	The jth grid cell, where $1 \leq j \leq X$.
P_{GCj}	Probability of the input sample to be in the jth grid cell.
$\hat{gc}$	Estimated grid cell.
GC	The set contains all grid cells.
$P(gc\|F)$	Probability of receiving a signal vector F at grid cell $gc \in GC$.
$\hat{l}$	Estimated user location.

to transmit data. In general, CQI is calculated based on various metrics such as signal-to-noise ratio (SNR), channel quality, and error rates. The CQI value is then used to adjust the modulation and coding scheme (MCS) for data transmission to optimize the transmission rate and reliability. The CQI values range from 0 to 15, with higher values indicating better channel quality.

B. FEATURES SELECTION

Among all LTE features, we used the twelve features introduced in DeepFeat [35]. They are the inputs to our model, as defined below:

- PCI, RSRP, RS SINR, RSRQ, CQI for serving cell: 5 features.
- PCI, RSRP for 3 neighbor Cell: 6 features.
- RSRQ for the 1st neighbor cell: 1 feature.

IV. SYSTEM MODEL

In this section, we present the SeqOutLoc and BiOutLoc system details. We will introduce the offline training phase, followed by the online localization phase. Figure 1 shows the block diagram of either SeqOutLoc or BiOutLoc models and Table 3 shows different notations that are used in this section.

A. OFFLINE TRAINING PHASE

The system model of our sequence-based framework using either LSTM or Bi-LSTM networks is trained during the offline training phase after collecting/wrangling the data, feature selection module, data augmentation, and generating the grid.

1) DATA COLLECTION

Data is collected using the TEMS Paragon tool [38]. TEMS Paragon is a solution offered by InfoVista that is used to measure and analyze the performance of wireless networks. It uses commercial smartphones to simulate real customer experience. Here we used data from Samsung Galaxy S8 mobile. The data is collected in real-time as the test equipment is driven or carried through the area of interest. TEMS Drive Test collects a wide range of parameters, including signal strength, signal quality, data rates, latency, and other key performance indicators. TEMS Drive Test is commonly used by network operators and service providers to optimize and troubleshoot their networks, assess the quality of service experienced by their customers, and plan and design future network deployments. It is also used by equipment vendors, regulatory agencies, and other organizations to evaluate the performance of wireless networks.

TABLE 4. Giza versus Sohag areas' characteristics.

Data Dimension	Giza Governorate	Sohag Governorate
Area name	Sheikh Zayed city	Sohag city
Area Categorization	Urban	Rural
Area size	45km^2	16km^2
Number of PCIs	204	114
Driving Distance	83km	21km
Data Samples	29375	7397

In our model, we use two datasets in Giza and Sohag governorates in Egypt with different area characteristics listed in Table 4. Sheikh Zayed city in Giza is mainly used in the training stage and benchmarking between the three models: FeedForward NN, LSTM, and BiLSTM. Sohag city was used in the transfer learning model, where we used the model's weights trained in Sheikh Zayed city as the initialization for Sohag city. The data is collected from the monthly usual drive test conducted from Etisalat by e& company in Egypt. Table 4 provides a detailed comparison between the two areas of interest.

Figure 2 and Figure 3 show the drive test route in a coloured theme, representing the range of RSRP values.

2) DATA AUGMENTATION

Data augmentation is the process of generating additional data from existing data to improve the performance of a machine-learning model [39]. In the context of outdoor localization, data augmentation can be used to increase the size and diversity of the training data set, which can help the model to generalize better to different environments and conditions. Augmentation also deals with the relatively low count of drive test samples in comparison to the training requirements of deep learning data, aiming to prevent overfitting. Overfitting occurs when the model is excessively tailored to the training set and struggles to perform effectively on unseen data. There are different types of data augmentation that can be applied to outdoor localization data, including the following:

1) Data synthesis: This involves generating new data points using algorithms or other methods, such as interpolation or extrapolation.

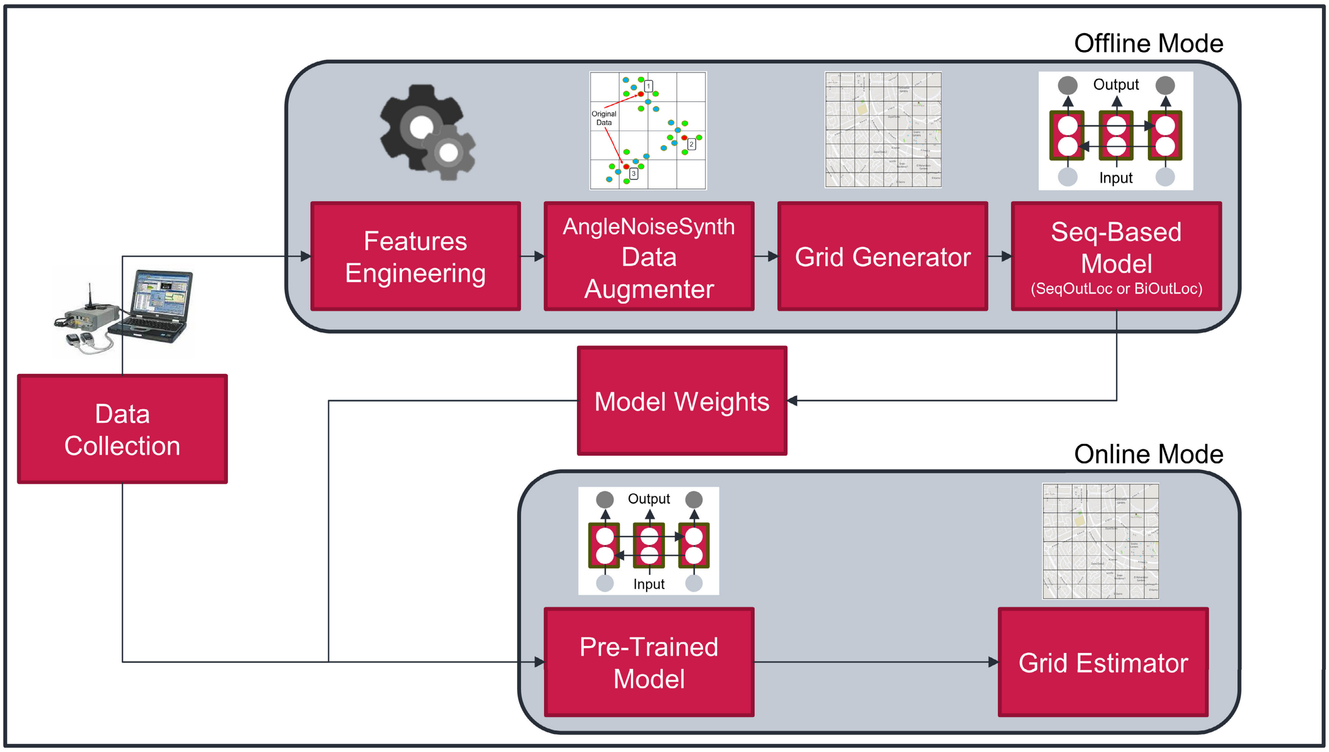

FIGURE 1. **System block diagram consists of: 1) data collection and wrangling; 2) offline mode for features engineering, data augmentation, grid generation, and model training; and 3) online mode, which uses a pre-trained model to estimate the location from new unseen data.**

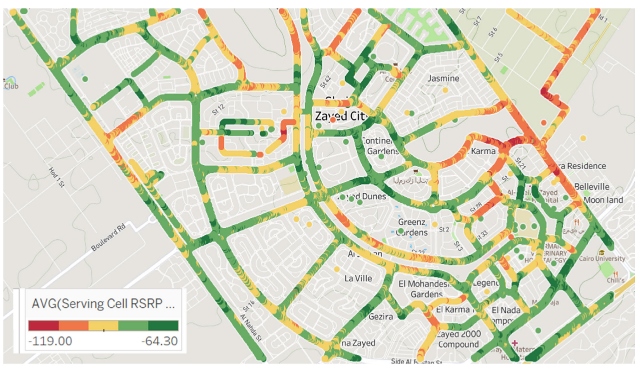

FIGURE 2. **Sheikh Zayed city (urban area) drive test route in a coloured theme, representing the range of RSRP values. Driving distance is 83 km, covering an area size of 45 km^2.**

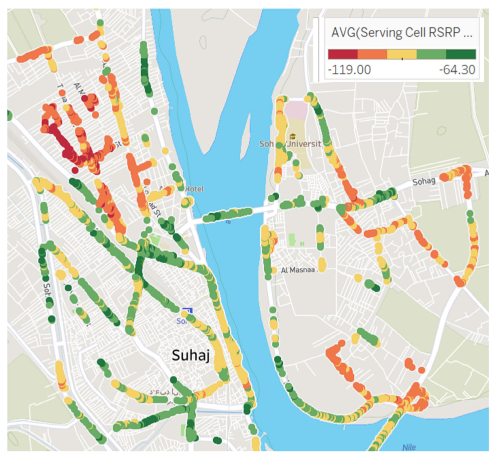

FIGURE 3. **Sohag city (rural area) drive test route. Driving distance is 21 km, covering an area size of 16 km^2.**

2) Noise injection: This involves adding noise to the data points to simulate the effects of measurement error, sensor sensitivity, or other sources of uncertainty. Generally, GPS has inherent errors that can range within a few meters, in some cases, up to tens of meters [40].
3) Data sampling: This involves sampling the data points at different rates or intervals, which can help the model to learn more robust representations of the data.

It is important to note that data augmentation should be carefully designed and evaluated to ensure that it does not introduce unrealistic or biased data into the training set. Data augmentation can be an effective tool to improve the performance of outdoor localization models, but it should be used in conjunction with other techniques, such as feature engineering and model selection. In our model, we introduced a new data augmenter that combines all the above factors in one model called AngleNoiseSynth Augmenter. The naming is coming that we used all dimensions illustrated above to build our augmenter as follows:

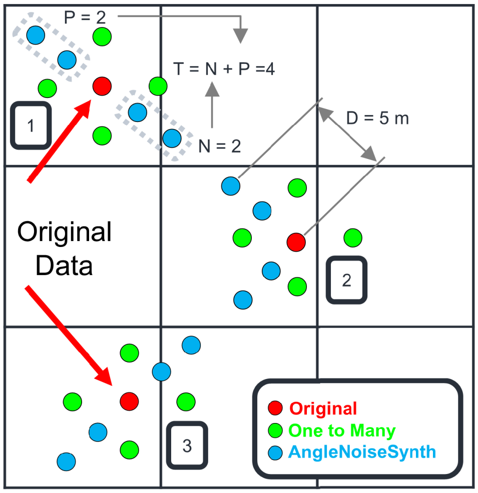

FIGURE 4. Illustration of the spatial augmentation with (AngleNoiseSynth) versus (One to Many) introduced in DeepFeat [35]. AngleNoiseSynth generates new $T = 4$ data points with $P = 2$ samples in the direction of the angle towards the previous sample and $N = 2$ samples in the direction of the next sample under the condition that all of them are within a maximum distance of $D = 5$ m.

1) Data synthesis: New T data points are generated according to the angle of movement of the previous and next data samples as shown in Figure 4. The algorithm will generate P samples in the direction of the angle towards the previous sample and another N samples in the direction to the next sample. In our model we used symmetric augmentation, where $P = N = 2$ and $T = P + N = 4$.
2) Noise Injection: Using the original data set, we determined the noise of the system per feature by monitoring the feature values variations over time when the mobile device is stationary. During these periods, the fluctuations observed can be attributed to noise in the system. By analyzing the amplitude of these fluctuations, an estimate of the noise level per feature can be obtained. By subtracting the averaged feature value from the measured value, the remaining variation can be attributed to noise. Finally, the noise is added to the additional data points generated in the data synthesis stage, simulating the effect of system noise, measurement error, or other sources of uncertainty.
3) Data Sampling: Each new sample is repeated under the condition that all of them will be within the maximum distance D meters. In our proposed models, we use $D = 5$ m, as GPS accuracy is negatively impacted after nearly 5 meters [41].

3) GRID GENERATION

The outdoor localization problem can be solved using two approaches. The first one is through formulating the problem as a regression problem that requires a massive amount of data training samples, increasing the complexity, and lowering the performance of the system. The second approach is to consider the problem as a classification problem using a grid approach [42]. Grid generation in outdoor localization refers to the process of dividing an area into a grid of cells, which can be used to represent the location of a device or to perform spatial operations such as interpolation or extrapolation. Grid generation is often used in outdoor localization to improve the accuracy and efficiency of location estimation. By dividing the area into a grid, it is possible to represent the location of a device more precisely and to incorporate spatial information into the localization algorithm. There are several factors that can influence the design of the grid in outdoor localization, including the size and shape of the area, the resolution of the grid, and the accuracy and availability of the location data. Grid generation can be useful for outdoor localization in various scenarios, such as when the location data is sparse or noisy, or when the location of the device needs to be estimated over a large area or at high resolutions. The grid generator divides the testing area into a virtual grid of X grid cells. One grid cell consists of all training samples with coordinates belonging to this cell and this represents the fingerprint of this cell. In addition, the grid cell size is a hyperparameter that needs to be tuned to maximize the localization accuracy, as will be shown in the performance evaluation section.

4) DEEP LEARNING MODELS

In our paper, we will compare between three different network models:

1) DeepFeat: is a robust deep learning framework designed for outdoor localization in LTE networks, as discussed in [35]. It employs a feedforward neural network model as its foundational deep learning architecture. Operating on the mobile operator side, DeepFeat utilizes an extensive feature set and diverse metrics from the mobile network to attain superior accuracy in localization.
2) SeqOutLoc: is our new proposed LSTM model. LSTM (Long Short-Term Memory) is a type of recurrent neural network (RNN) that is commonly used for tasks such as natural language processing, speech recognition, and time series prediction [43]. It is well-suited for these tasks because it is able to capture long-term dependencies in the data by using memory cells and gates that control the flow of information through the network [44]. In the context of outdoor localization, LSTM models can be used to improve the accuracy of location estimation by incorporating temporal information into the model. For example, the movements and behaviors of a device can be modeled over time using an LSTM model, which can help to predict the device's future location and to filter out noise and errors in the location data. LSTM models can be particularly useful for outdoor localization in scenarios where the location data is noisy or sparse, or where the device is moving in a complex or

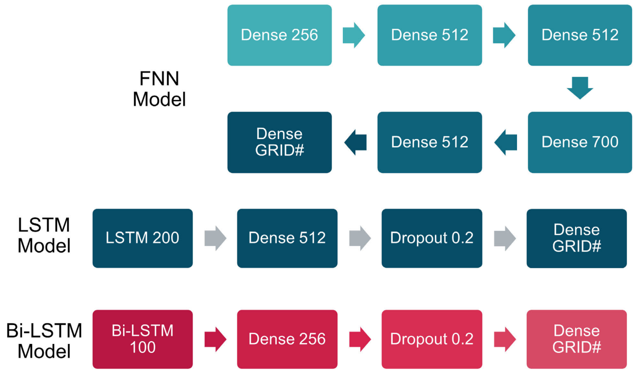

FIGURE 5. FNN, LSTM, and Bi-LSTM models block diagram and main parameters' values.

unpredictable manner [45]. This was clear in the results where the LSTM model shows higher immunity against noisy samples than FNN one.

3) BiOutLoc: is our second proposed model that uses Bi-LSTM network. Bi-LSTM (Bidirectional Long Short-Term Memory) is a type of recurrent neural network (RNN) that is similar to a regular LSTM model, but it processes the input data in both forward and backward directions. This allows the Bi-LSTM model to capture dependencies in the data that are both past and future relative to the current time step. In a Bi-LSTM model, the input data is processed by two separate LSTM networks: one that processes the data in the forward direction and one that processes the data in the backward direction. The output of these two networks is then combined to produce the final output of the model [46], [47]. Three models with their parameters are shown in Figure 5. In all models, we used the Dense block, which refers to a dense hidden layer where each neuron is connected to every neuron in the previous layer, thus being called dense. This layer is the most commonly used layer in artificial neural network networks. The number following the Dense block refers to the number of neurons or units in that specific layer.

The input of the three models is the J used features $(F_1, F_2, \ldots, F_J)$, where $J = 12$ resulting from the feature selection module. We have Ls training samples representing the augmented training data set, where each record in the Ls samples contains all the J input features. The models' output is the probability that the input sample belongs to each grid cell $(P_{gc1}, P_{gc2}, \ldots, P_{gcX})$.

Finally, model uses Softmax as an output layer. Softmax is a type of activation function that transforms the output of the model into a probability distribution, which can be used to classify the input data into different grid cells. Softmax is perfect in the context of multi-class classification, where the output of the model is a probability distribution over a set of classes. It is often used in conjunction with a cross-entropy loss function to train.

TABLE 5. Sequence-based model hyper-parameters values.

Hyper-parameter	Value
Input Dimension	Number of Features
Final Layer	Softmax, Grids size
Loss function	categorical crossentropy
Learning Rate	0.001
Optimizer	ADAM
Activation function	Relu
Validation data%	20%
Testing data%	20%
Batch size	2048
Drop Out	0.2

We use a laptop with a processor Intel(R) Core(TM) i7-10750H CPU @ 2.60GHz and dedicated 8 GB GPU accelerator and 16 GB of RAM and equipped with a 475G SSD hard disk on windows 10 64-bit. We built the model using the TensorFlow framework.

Sequence-based model parameters used during the training phase are listed in Table 5. We intended to use 20% of total samples as validation data to avail more instances for evaluating the model's performance, avoid overfitting, and help in selecting the best-performing model. We reserved 20% of the total data set as well in the testing phase to ensure a highly reliable assessment of how well the trained model generalizes to new, unseen data in real-world performance.

B. ONLINE PHASE

Here we use the trained model in the offline mode with all parameters weights after the training phase, then we provide the new samples to the model to predict related grid cells and estimate its localization accuracy.

1) USE LOCATION ESTIMATION

Our trained model generates the most likelihood grid cell that UE should belong to. First, we can estimate the grid cell $(\hat{gc})$ where the user exists, which is the cell with maximum probability, as following:

$$\hat{gc} = \underset{gc \in GC}{\arg\max}(P(gc|F)). \tag{3}$$

where GC is the set containing all grid cells, gc is any grid cell belong to GC, and $F = (F_1, F_2, \ldots, F_J)$ is the input features vector. We consequently estimate the user's location $\hat{l}$. However, it is a poor estimation. We used a second method with better estimation by calculating user location $\hat{l}$ based on the weighted centroid. The weighted centroid is the center of mass of all grid cells weighted by the corresponding probability of each cell [5].

$$\hat{l} = \sum_{gc \in GC} gc \times P_{gc}. \tag{4}$$

where P_{gc} is the corresponding probability for each grid cell gc at the deep model's output layer.

2) LOCALIZATION ACCURACY

We calculate the system accuracy through two metrics:

- The median absolute error (MAE) between the actual samples location and the estimated one from the online phase. The lower the better.
- The interquartile range (IQR) which is a measure of dispersion that is commonly used in statistics and data analysis. It is defined as the difference between the third quartile (75th percentile) and the first quartile (25th percentile) of a data set. The IQR is a robust measure of dispersion that is less sensitive to outliers than other measures such as the range or standard deviation. We mainly used it in the comparison section to represent our model robustness against DeepFeat model.

V. PERFORMANCE EVALUATION

In this section, we evaluate the performance of both the SeqOutLoc and BiOutLoc models in comparison to the feed forward model introduced in DeepFeat. Our benchmarking was conducted over an urban environment (Sheikh Zayed city). We study the effect of changing different parameters on system performance. Finally, we demonstrate the improvement in accuracy achieved through the use of transfer learning techniques in a different area (Sohag city).

A. EFFECT OF HYPER-PARAMETERS TUNING ON LOCALIZATION ACCURACY:

In this section, we will present the results of our work. We used the same dataset used in DeepFeat in Sheikh Zayed city in Giza governorate and the Neural network model was our reference model in the benchmark where we compared it with our new LSTM and Bi-LSTM models. Then we study the effect of different parameters on SeqOutLoc and BiOutLoc performance. Finally, we compare SeqOutLoc and BiOutLoc accuracy and robustness with DeepFeat.

1) DATA AUGMENTATION

The Figure 6 shows the effect of using One to Many presented in DeepFeat [35] versus our proposed data augmenter AngleNoiseSynth over same neural network model used in DeepFeat. Both are compared with original data without a data augmenter. Using 200 epochs, our new data augmenter shows a better median absolute error = 15.9 m compared to 16.9 m using One to Many, which is a 5.9% enhancement. The clear enhancement is shown in the IQR metric, where AngleNoiseSynth superseded One to many by 36%. This shows how the new data augmentation technique enhanced system robustness.

2) NUMBER OF EPOCHS

An epoch is a complete pass of the training data through a machine learning (ML) model during the training process.

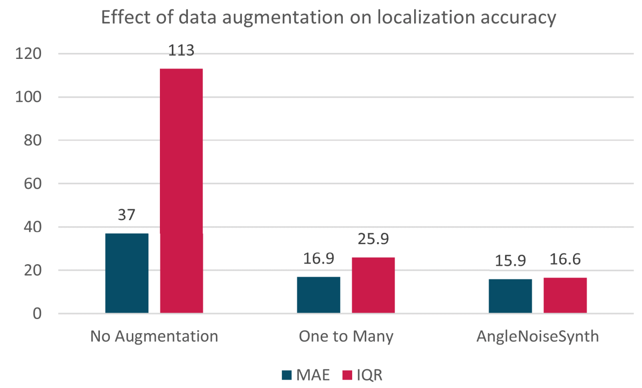

FIGURE 6. Effect of data augmentation techniques on localization accuracy and system robustness versus no augmentation: AngleNoiseSynth outperforms in both MAE and IQR.

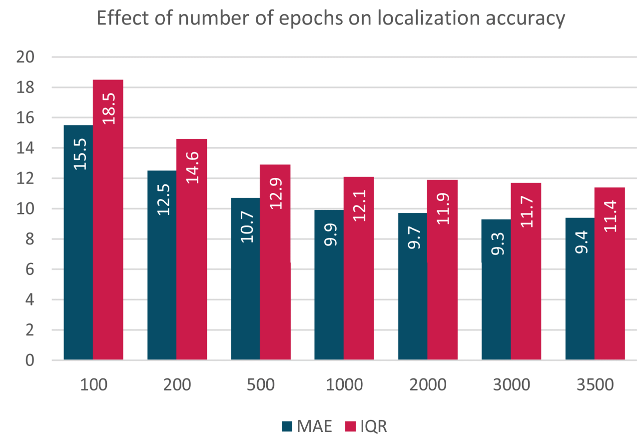

FIGURE 7. Effect of number of epochs on localization accuracy. Accuracy increases as the number of epochs increases, but begins to saturate after 2000 epochs.

An epoch consists of one or more batches of training data, which are processed by the model in sequence. At the end of an epoch, the model's parameters are updated based on the error or loss calculated on the training data. The Figure 7 shows the effect of changing the number of training epochs on the localization accuracy at 50m grid cell length using the Bi-LSTM model. Localization accuracy is increasing with a higher number of epochs. However, the localization accuracy is nearly constant after 2000 epochs. In our final version of the architecture, we used 3500 epochs, where MAE equals 9.4m and IQR equals 11.4m.

3) BATCH SIZE

The batch size determines the number of training samples utilized in one iteration during optimization. The choice of batch size can have various effects on the training process and the performance of the model. Table 6 shows the effect of changing batch size on the localization accuracy and training time at 50m grid cell length using the Bi-LSTM model with 100 epochs. Localization accuracy shows optimum values of 13.1 and 13.2 m with 1024 and 2048 batch sizes, respectively.

TABLE 6. Effect of different batch sizes on localization accuracy and 1 epoch training time in sec. A higher size shows lower training time, where 2048 demonstrates the balanced optimal value for MAE, IQR, and training time.

Metric	Batch size				
	256	512	1024	2048	4096
Training Time	36.2	23.6	15.6	10.2	7.8
MAE	14.2	13.7	13.1	13.2	14.0
IQR	17.6	17.5	17.1	15.0	18.7

TABLE 7. Effect of different dropout values on localization accuracy. A dropout of 0.2 shows the optimal value for MAE and IQR.

Accuracy	Dropout value					
	No dropout	0.1	0.2	0.3	0.4	0.5
MEA	14.2	12.8	12.2	12.2	12.3	12.5
IQR	17.7	14.7	13.8	13.9	14.1	15.4

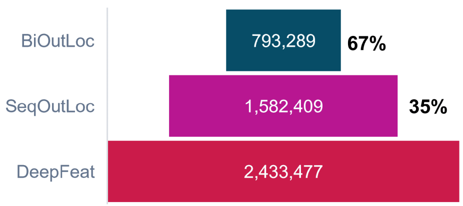

FIGURE 8. Count of parameters per model. The percentage shows the reduction in each model's parameters compared to DeepFeat.

However, 2048 has a better IQR of 15 m as well as less training time. In our final version of the architecture, we used a batch size of 2048.

4) DROPOUT VALUE

Dropout is a regularization technique that randomly drops a fraction of input units during training, preventing overfitting. Table 7 shows the effect of changing the dropout value on the localization accuracy. Localization accuracy shows optimum values of 12.2 m with a 0.2 dropout value. As stated earlier, we used a dropout value of 0.2 in our final version of the architecture.

B. PERFORMANCE BENCHMARK

we evaluate the performance of SeqOutLoc and BiOutLoc versus DeepFeat in an urban environment (Sheikh Zayed city in Giza).

1) NUMBER OF MODEL PARAMETERS

it is often desirable to decrease the number of parameters in the model for better computational efficiency, which is important in localization applications where the model is used in real-time. By decreasing the number of parameters, it is possible to reduce the risk of over-fitting and improve the generalization performance of the model. It is important to note that there is a trade-off between model complexity and performance.

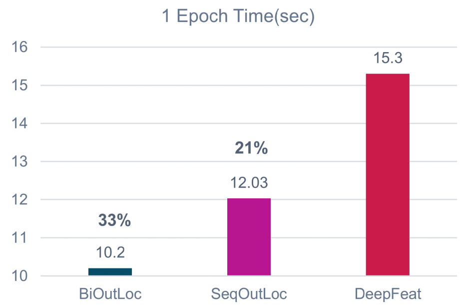

FIGURE 9. Epoch time per model. The percentage shows the reduction in each model's 1 epoch time compared to DeepFeat.

In this study, we evaluated the performance of our two proposed models, SeqOutLoc and BiOutLoc, against the DeepFeat model in an urban environment. We found that while DeepFeat required 2.433 million parameters to achieve its best accuracy, SeqOutLoc achieved an improvement of 35% with only 1.58 million parameters. Additionally, BiOutLoc demonstrated the most significant reduction in parameters, utilizing only 793.3K and outperforming the FNN model by a 67% reduction in memory allocation. These results highlight the efficiency of using sequential data in outdoor localization, as opposed to single-point-based models, which require additional resources to compensate for the lack of temporal information. For the enhancement seen in BiOutLoc compared to SeqOutLoc, we utilized the information that data training in a bidirectional manner is better than the unidirectional one because it uses both previous data and future data. This extra information encourages us to minimize the dense layer in SeqOutLoc from 512 to 256 in BiOutLoc. This resulted in a 49.8% reduction in the number of parameters between the two models with better localization accuracy in BiOutLoc.

2) TRAINING TIME

In this study, we utilized the epoch time to evaluate the overall training duration for the three models under consideration. As depicted in Figure 9, both SeqOutLoc and BiOutLoc models outperform DeepFeat by 21% and 33% respectively. Specifically, DeepFeat completes one epoch in 15.3 seconds, SeqOutLoc in 12.03 seconds, and BiOutLoc in 10.2 seconds. These results demonstrate the efficiency of our proposed models in terms of training time.

C. COMPARATIVE STUDY

In this study, we include different dimensions to evaluate the performance of our newly introduced models, SeqOutLoc and BiOutLoc, against the DeepFeat model using a common dataset.

- We compare AngleNoiseSynth data augmentation technique to the One to Many method utilized in the DeepFeat model
- We compare using two different number of epochs 200 and 3500 to assess model conversion

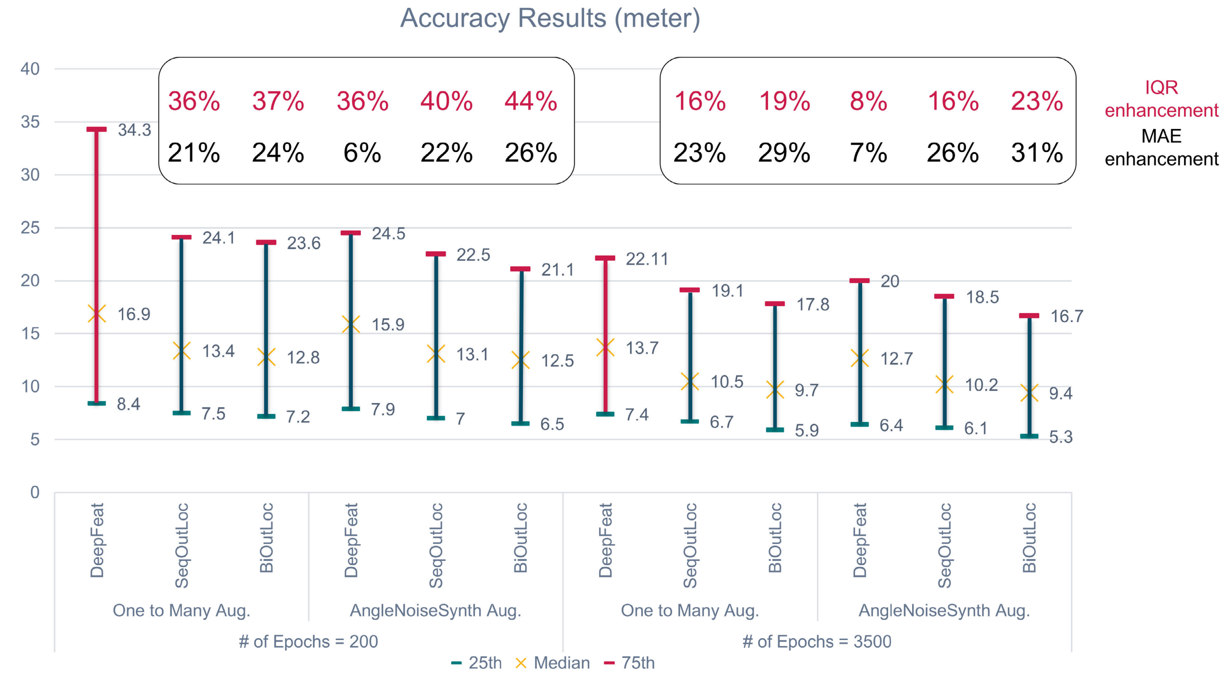

FIGURE 10. **Performance comparison: SeqOutLoc and BiOutLoc vs. DeepFeat model, assessing augmentation technique impact across epochs (200 and 3500). DeepFeat with One to Many augmenter (highlighted in red) is our reference in calculating MAE and IQR enhancement %.**

- We use median absolute error (MAE) and interquartile range (IQR) as our benchmark metrics.
- We use DeepFeat with One to Many augmenter result as our reference value for both epochs number.

Figure 10 illustrates the output of this comparison with below main observations:

- BiOutLoc with AngleNoiseSynth supersedes DeepFeat best results with epchos=3500. MAE enhanced by 31% (from 13.7 to 9.4 m) and IQR enhanced 23% (from 14.7 to 11.4 m).
- We see how new data augmenter AngleNoiseSynth achieves 36% enhancement in IQR compared to One to Many (16.6m compared to 25.9m) using DeepFeat with low number of epochs (200). This shows how new augmenter fastens the model convergence with more robust system.
- Any sequence-based model (SeqOutLoc and BiOutLoc)shows better MAE compared to FNN model by at least 18%.

D. TRANSFER LEARNING

Transfer learning is an improved learning technique that utilizes the knowledge from an existing task that has already been learned for a new task. It minimizes the training processing and provides better accuracy as it converges to the optimum weights better than the old task and this achieves better accuracy. In our work, we initialize a new Bi-LSTM model for Sohag city from the final weights extracted from the learning process of Sheikh Zayed city and use them for the prediction of Sohag city. Using BiOutLoc with 3500 epochs, accuracy was enhanced by 18%, where MAE dropped from 6.5 without transfer learning to 5.3 m with it. Training time was reduced by 71% from 4.3 to 1.25 hours, as stated in Table 8.

TABLE 8. **Effect of transfer learning of the final weights of the Sheikh Zayed city model to initialise the new Bi-LSTM model for Sohag city on localization accuracy and training time. Using 3500 epochs, accuracy enhanced by 18% and training time by 71%.**

State	Median Absolute Error	Training Time
Without Transfer Learning	6.5m	4.3 hours
With Transfer Learning	5.3m	1.25 hours

VI. CONCLUSION

In this paper, we introduce two sequence-based frameworks with significant accuracy enhancements in large-scale outdoor environments. The first uses a uni-directional LSTM network called SeqOutLoc, and the second uses a bi-directional LSTM network called BiOutLoc. In both frameworks, we introduced AngleNoiseSynth augmenter to extend the dataset and enhance the robustness of the systems based on the angle of user movement and system noise. For SeqOutLoc, we show how adding sequence information enhances the accuracy of localization in a large-scale outdoor urban area of 45 km^2 by 25% compared to previous work, while using 35% fewer network parameters.

The second model, BiOutLoc, enhances the localization accuracy with fewer network parameters by utilizing both past and future information, which is useful in near-real-time localization. To the best of our knowledge, our work is the first to use a Bi-LSTM model in outdoor fingerprint-based localization. BiOutLoc achieves a median localization accuracy of 9.4 meters, surpassing other deep learning-based localization systems by 31%, while reducing the number of parameters by 67%. Finally, we apply the transfer learning technique by initializing a new model with a different dataset using the final weights extracted from the original model. This resulted in an 18% enhancement in localization accuracy and a 71% reduction in training time compared to results without transfer learning.

The current proposed models depend on data collected from the field from one device model. It is recommended to investigate the models' performance for different devices used for data collection for future research. Moreover, we can extend our work to cover other urban areas without a transfer learning phase (i.e., by adding more features to the model to replace the transfer learning phase). Finally, we can test our work for new, evolving mobile technologies like 5G.

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